DAY 22 | POST-READ

Context and RAG: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Context engineering	Deciding what goes into the context: instructions, input, history, knowledge, tools, examples.
Principles	Relevant over complete, budget tokens, order matters, label sources, compress history.
RAG	Retrieve relevant chunks and generate an answer grounded in them, with citations.
Ingestion pipeline	Load, clean, chunk, embed, store with metadata.
Query pipeline	Question, retrieve top-k, assemble, generate, cite.
Grounded prompt	Use only the context; say "I don't know"; cite chunk ids; low temperature.
Chunking	Fixed, sentence or paragraph, recursive, structure aware, semantic.
Chunk size	Start around 300 to 800 tokens, 10 to 20 percent overlap, then measure.

Part B: Going beyond today

1. Lost in the middle

Studies of long-context models have found that information placed in the middle of a long context is used less reliably than information near the start or end. Put the most relevant chunks first, and keep context lean.

2. Parent-child chunking

Index small chunks for precise matching, but send the larger parent section to the model so it has full context. Many RAG frameworks support this pattern.

3. Contextual chunk headers

```
chunk_text = f"Document: Employee Handbook > Section: Leave policy\n{chunk}"
```

Prefixing each chunk with its document and section titles helps both retrieval and the model's understanding.

4. Conversation memory

Keep the last few turns verbatim and a running summary of older turns. Store long-term facts ("prefers answers in Japanese") separately and add them only when relevant.

5. When not to use RAG

If the knowledge is small and stable, put it straight into the prompt. If the task is about style or format rather than facts, examples or fine-tuning may suit better. RAG shines for large, changing knowledge.

Part C: Check yourself

- Name five things that can go into a context window.
- What is the difference between the ingestion and query pipelines?
- What should a grounded prompt tell the model to do when the answer is missing?
- What goes wrong with chunks that are too small? Too large?
- Why store metadata with each chunk?

Part D: Practice ideas for tonight

- Chunk a real policy document from your work (non-confidential) three ways and compare the chunks.
- Add contextual section headers to the notebook's chunker and re-test retrieval.
- Write a grounded prompt that also asks the model to quote the exact sentence it used.

Coming up on Day 23: embeddings, vector databases, approximate nearest neighbour search, hybrid search and re-ranking.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today